

LECTURE SHEET

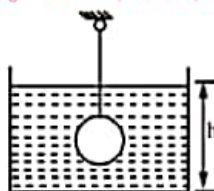
EXERCISE-I

(Pressure in a Fluid, Pascal's Law, Atmospheric pressure and Barometer Archimede's principle)

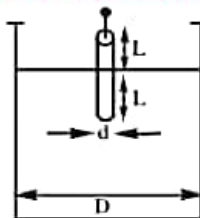
LEVEL-I (MAIN)

Straight Objective Type Questions

1. A metal sphere connected by a string is dipped in a liquid of density ρ as shown in figure. The pressure at the bottom of the vessel will be, (p_0 = atmospheric pressure)

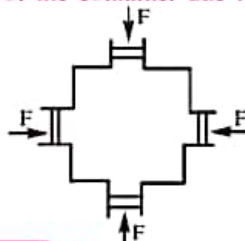


- 1) $p = p_0 + \rho gh$ 2) $p > p_0 + \rho gh$ 3) $p < p_0 + \rho gh$ 4) p_0
2. A sphere of solid material of specific gravity 8 has a concentric spherical cavity and just sinks in water. Then, the ratio of the radius of the cavity to the outer radius of the sphere must be
- 1) $\frac{\sqrt{3}}{2}$ 2) $\frac{\sqrt{5}}{2}$ 3) $\frac{\sqrt{7}}{2}$ 4) $\frac{2}{\sqrt{7}}$
3. An open U-tube contains mercury. When 11.2 cm of water is poured into one of the arms of the tube, how high does the mercury rise in the other arm from its initial level?
- 1) 0.82 cm 2) 1.35 cm 3) 0.41 cm 4) 2.32 cm
4. A candle of diameter d is floating on a liquid in a cylindrical container of diameter D ($D \gg d$) as shown in figure. If it is burning at the rate of 2cm/h, then the top of the candle will



- 1) remain at the same height 2) fall at the rate 1 cm/h
- 3) fall at the rate of 2 cm/h 4) go up at the rate of 1 cm/h
5. In the figure shown water is filled in a symmetrical container. Four pistons of equal area A are used at the four opening to keep the water in equilibrium. Now an additional force F is applied at each piston. The increase in the pressure at the centre of the container due to this addition is

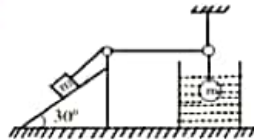
- 1) $\frac{F}{A}$
- 2) $\frac{2F}{A}$
- 3) $\frac{4F}{A}$ 4) 0



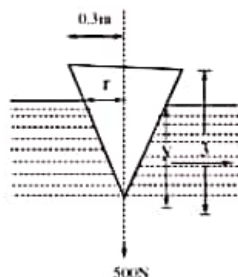
LEVEL-II (ADVANCED)

Straight Objective Type Questions

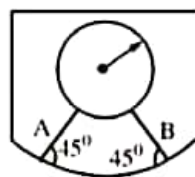
1. A block of mass 'm' is kept over a fixed smooth wedge. The block is attached to a sphere of same mass as shown in figure, which is immersed in water. If specific gravity of material is 2 then the acceleration of sphere is



- a) zero b) 2m/s^2 c) 2.5 m/s^2 d) 5 m/s^2
2. A light circular cone buoy is pressed by a 500N force as shown in figure then the depth upto which it will submerge is



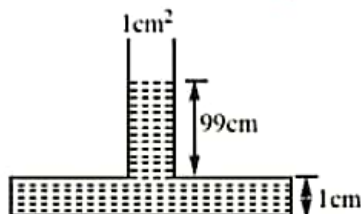
- a) 1.7 m b) 2.0 m c) 2.1 m d) 3.0 m
3. The density of ice is $x\text{ gm/cc}$ and that of water is $y\text{ gm/cc}$. what is the change in volume in cc, when $m\text{ gm}$ of ice melts ?
- a) $M(y-x)$ b) $(y-x)/m$ c) $mxy (x-y)$ d) $m(1/y - 1/x)$
4. A hollow sphere of mass 50kg and radius $\left(\frac{3}{40\pi}\right)^{1/3}\text{ m}$ is immersed in a tank of water of density 10^3 kg/m^3 . The sphere is tied to the bottom of the tank by two wires A and B as shown and is in equilibrium then. Tension in wire A is ($g = 10\text{m/s}^2$)



- a) $125\sqrt{2}\text{N}$ b) $250\sqrt{2}\text{N}$ c) 250 N d) 125 N
5. An inverted u-tube has its two limbs in water and kerosene contained in two beakers. If water rises to a height of 10cm , to what height does kerosene (density= 0.8gm/cc) rise in the other limb?
- a) 10cm b) 12.5cm c) 15cm d) 20cm
6. If the atmospheric pressure is 76cm of mercury, at what depth of water the pressure will becomes 4 atmospheres ? (nearly)
- a) 31m b) 21m c) 11m d) 9

7. A piston of cross-sectional area 100 cm^2 is used in a hydraulic press to exert a force of 10^7 dyne on water. The cross-sectional area of the other piston which supports a truck of mass 200 kg is,
 a) 1.96 cm^2 b) 0.196 m^2 c) 0.98 m^2 d) 98 m^2
8. A tube 1 cm^2 in cross-section is attached to the top of a vessel 1 cm high and of cross section 100 cm^2 . Water is poured into the system filling it to a depth of 100 cm above the bottom of the vessel as shown in figure then find force exerted by water against the bottom of the vessel ?

- a) 100 N
 b) 200 N
 c) 300 N
 d) 400 N

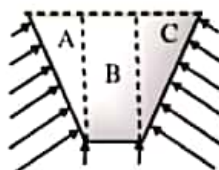


More than One correct answer Type Questions

9. A liquid is filled up to height h in a vessel, as shown. Find correct option (s):



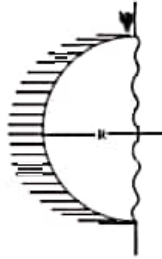
- a) If $\alpha = \beta$, horizontal component of forces on left and right side on inclined faces will be equal and opposite.
 b) If $\alpha \neq \beta$ horizontal component of forces on left and right side of inclined faces will be equal and opposite.
 c) If A is the area of the base of the vessel, then force exerted by liquid on the walls of vessel is greater than $(P_{\text{atm}} + \rho gh)A$
 d) As above, the force exerted by liquid on walls is equal to $(P_{\text{atm}} + \rho gh)A$
10. In the following figure shown, a liquid is filled into the vessel up to a height h and the forces exerted against the liquid by the walls of the vessel are shown by arrows, the force being everywhere perpendicular to the walls of the vessel. For this situation, mark out the correct statement(s)



- a) The horizontal component of the forces on the left and right side inclined faces will cancel each other
 b) The weight of the section lettered is supported by vertical component of forces acting on the inclined faces
 c) Force exerted by the liquid on the base of the vessel is greater than $(P_0 + \rho gh)A$ where ρ is the density of the liquid and A is base area of the vessel
 d) Net force exerted by liquid on the base of the vessel is in the vertical downward direction

Linked Comprehension Type Questions**Passage - I :**

A tortoise is just sinking in water of density ρ . The tortoise is assumed to be a hemisphere of radius R .



11. Find the horizontal thrust

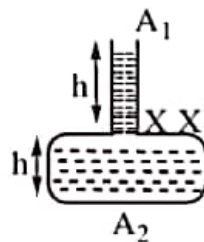
- a) $\rho g \pi R^3$ b) $\frac{1}{3} \rho g \pi R^3$ c) $\frac{2}{3} \rho g \pi R^3$ d) 0

12. Find Vertical thrust

- a) $\rho g \pi R^3$ b) $\frac{1}{3} \rho g \pi R^3$ c) $\frac{2}{3} \rho g \pi R^3$ d) 0

Matrix Matching Type Questions

13. The vessel shown in the fig. has two sections of area of cross – section A_1 and A_2 . A liquid of density fills both the sections, up to a height h in each and is in static condition. Neglect atmospheric pressure.

**Column-I**

- A) The net force exerted by the liquid on all side walls is
 B) The force exerted by the liquid on the bottom of the vessel is
 C) The weight of the liquid contained in the vessel is
 D) The horizontal walls of the liquid vessel at the level xx' exert a downward force which is

Column-II

- P) $2h\rho g A_2$
 Q) Zero
 R) $h\rho g(A_2 - A_1)$
 S) Less than $2h\rho g A_2$

Integer Type Questions

14. A cuboidal piece of wood has dimensions a , b and c in S.I system of units. Its relative density is d which is numerically equal to $1/a$. It is floating in a large body of water such that side a is vertical. It is pushed down a bit and released. The time period of SHM executed by it is _____ seconds. (Assume $g = \pi^2 \text{ms}^{-2}$)
15. A sphere falls from rest into water from a height of 2 m. the relative density of the sphere is 0.80. find the depth to which ball will sink (in m)

EXERCISE-II

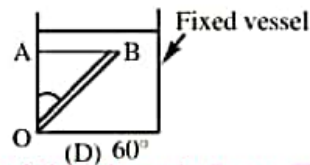
(Pressure Difference and Buoyant force in Accelerating Fluids, Flow of Fluids, Steady and Turbulent flow)

LEVEL-I (MAIN)

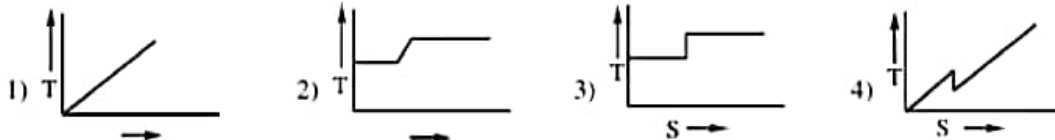
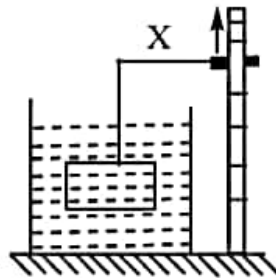
Straight Objective Type Questions

1. A uniform rod OB of length 1m, cross-sectional area 0.012 m^2 and relative density 2.0 is free to rotate about O in vertical plane. The rod is held with a horizontal string AB which can withstand a maximum tension of 45 N. The rod and string system is kept in water in equilibrium as shown in figure. The maximum value of angle α which the rod can make with vertical without breaking the string is

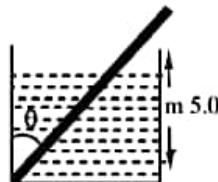
- 1) 45° 2) 37°
3) 53° 4) 60°



2. A metallic plate having shape of a square is suspended as shown in figure. The plate is made to dip in water such that level of water is well above that of the plate. The point X is then slowly raised at constant velocity then curve between tension T in string and displacement S of point X is give by



3. A wooden plank of length 1m and uniform cross section is hinged at one end to the bottom of a tank as shown. The tank is filled with water upto a height of 0.5m. The specific gravity of the plank is 0.5. The angle θ made by the plank in equilibrium position is



- 1) 30° 2) 45° 3) 60° 4) 90°

4. A cube of ice of side 4cm is placed in an empty cylindrical glass vessel of inner diameter 6cm. Assume that the ice melts uniformly from each side so that it always retains its cubical shape. Find the length of the edge of the ice cube at the instant it just leaves the contact with the bottom of the vessel.
- 1) 0.26 cm 2) 1.26 cm
3) 2.26 cm 4) 3.26 cm

LEVEL-II (ADVANCED)

Straight Objective Type Questions

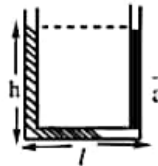
1. A U – tube of base length “ l ” filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration “ a ” on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero, then the height h is given by :

$$\text{a) } \frac{a}{2g} l$$

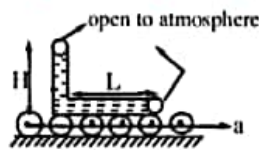
b) $\frac{3a}{2g}l$

c) $\frac{a}{a^2 - 2b}$

d) $\frac{2a}{3g}l$



2. A narrow tube completely filled with a liquid is lying on a series cylinders as shown in figure. Assuming no sliding between any surfaces, the value of acceleration of cylinders for which liquid will not come out of the tube from anywhere is given by

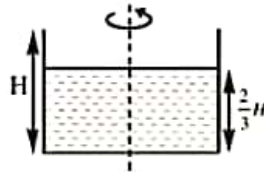


- a) $gH/2L$ b) gH/L c) $2gH/L$ d) $gH/\sqrt{2L}$
3. A cork is submerged in water by a spring attached to the bottom of a bowl. The the bowl is kept in an elevator moving with acceleration downwards, the length of spring
a) Increases b) Decreases c) Remains unchanged d) None of these
4. Two metals are mixed together in equal volumes to form an alloy of specific gravity 4. When equal masses of the same two metals are mixed together, the specific gravity of the alloy is 3, the specific gravity of each metal is
a) 4 and 6 b) 2 and 6 c) 2 and 4 d) 4 and 8
5. A balloon contains 400m^3 of hydrogen. If the mass of the balloon without hydrogen is 34kg . Its net lifting ability is (Density of hydrogen = 0.09 kg/m^3) Density of air = 1.3 kg/m^3)
a) 590kg.wt b) 450kg.wt c) 70kg.wt d) 36kg.wt
6. Density of ice is ρ and that of water is σ What will be the decrease in volume when a mass M of ice melts.
a) $\frac{M}{\sigma - \rho}$ b) $\frac{\sigma - \rho}{M}$ c) $M \left[\frac{1}{\rho} - \frac{1}{\sigma} \right]$ d) $\frac{1}{M} \left[\frac{1}{\rho} - \frac{1}{\sigma} \right]$

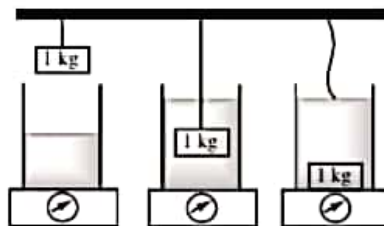
7. A large block of ice 5 m thick has a vertical hole drilled in it and is floating in a lake. the minimum length of the rope required to draw a bucketfull of water through the hole is (density of ice = 900 kg/m^3)
 a) 0.5 m b) 1m c) 4 m d) 4.5 m
8. A cubical block of wood of side 10 cm floats at the interface between an oil and water with its lower surface 2cm below the interface. the heights of the oil and water columns are 10 cm each. Density of oil is 0.8 g/c.c. . The mass of the block is
 a) 720 gram b) 900 gram c) 960 gram d) 840 gram
9. An ice berg of density 900 kg/m^3 is floating in water of density 1000 kg/m^3 . The percentage of volume of ice cube out side the water is
 a) 20% b) 35 % c) 10% d) 25%
10. A piece of copper having an internal cavity weighs 264 gm in air and 221gm in water. The volume of the cavity is (Density of copper = 8.8 gm/cc)
 a) 43cc b) 30cc c) 13cc d) 70cc

More than One correct answer Type Questions

11. A circular cylinder of radius R and height H is filled with water to a height $(2/3)H$. It starts rotating about its axis with constantly increasing angular speed. Choose the correct alternatives.



- a) At all speeds, shape of the free surface is paraboloid
 b) The free surface touches first the brim of the cylinder and then the base of the cylinder
 c) The free surface cannot touch the base without spilling water.
 d) The free surface touches the brim as well as the base at the same instant.
12. Three identical blocks each of mass $m = 1 \text{ kg}$ and volume $3 \times 10^{-4} \text{ m}^3$ are suspended by massless strings from a support as shown. Underneath are three identical containers containing the same amount of water that are placed over the scales in fig(a), the block is completely out of the water, in fig(b), the block is completely submerged but not touching the beaker and in fig(c), the block rests on the bottom of the beaker. The scale in fig(a) reads 14N.

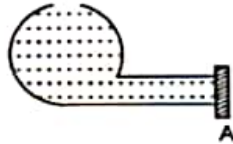


- a) The tension in the string in (b) is 10N. b) The tension in the string in (b) is 7N
 c) The reading of the scale in (b) is 17 N d) The reading of the scale in (c) is 24N

Integer Type Questions

13. The work done by pressure in forcing 2 m^3 of water through a 1cm of pipe, if the pressure difference across the pipe is 10^5 Pa is $n \times 10^5 \text{ J}$ where $n =$ _____

14. In a vessel as shown, the opening has a cross-sectional area A . F_1 is the net force applied on the plate by liquid and air, which is kept to close the opening. The plate is now displaced a short distance away from the opening in which case liquid strikes the plate inelastically with a force F_2 . Find F_2/F_1



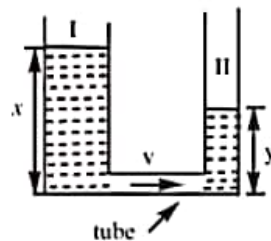
EXERCISE-III

(Irrorational flow of an incompressible and non viscous fluid, equation of continuity, Bernoulli's equation, applications of Bernoulli's equation)

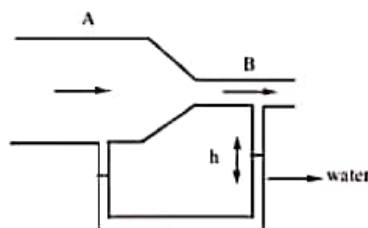
LEVEL-I (MAIN)

Straight Objective Type Questions

1. If cross-sectional area of limb I is A_1 and that of limb II is A_2 , then velocity of the liquid in the tube will be, (cross-sectional area of tube is very small)



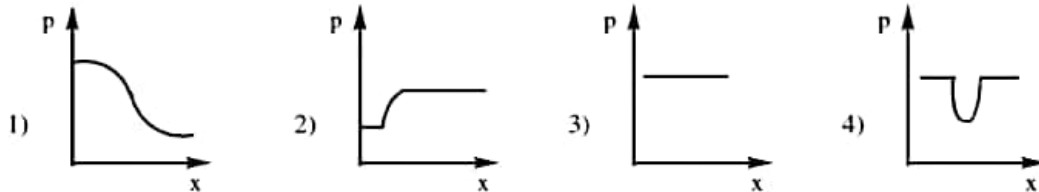
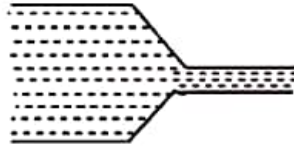
- 1) $\sqrt{2g(x-y)}$ 2) $\frac{A_1}{A_2} \sqrt{2g(x-y)}$ 3) $\frac{A_2}{A_1} \sqrt{2g(x-y)}$ 4) None of above
2. Air is blown through a pipe AB at a rate of 15 L/min. The cross-sectional area of the broad portion of the pipe AB is 2cm^2 and that of the narrow portion is 0.5cm^2 . The difference in water level h is (density of air = 1.32 kg/m^3)



- 1) 16 mm 2) 1.5 mm 3) 10 mm 4) 3.2 mm
3. A container has a small hole at its bottom. Area of cross-section of the hole is A_1 and that of the container is A_2 . Liquid is poured in the container at a constant rate $Q\text{m}^3/\text{s}$. The maximum level of liquid in the container will be

- 1) $\frac{Q^2}{2g A_1 A_2}$ 2) $\frac{Q^2}{2g A_1^2}$ 3) $\frac{Q}{2g A_1 A_2}$ 4) $\frac{Q^2}{2g A_2^2}$

4. Water flows through a frictionless duct with a cross-section varying as shown in figure. Pressure p at points along the axis is represented by



5. Water filled in an empty tank of area 10 through a tap of cross sectional area A . The speed of water flowing out of tap is given by $v(\text{m/s}) = 10 \left(1 - \sin \frac{\pi}{30} t \right)$ where ' t ' is in seconds. The height of water level from the bottom of the tank at $t = 15$ sec will be

- 1) 10 m 2) $15 + \frac{30}{\pi}$ m 3) $\frac{5}{4}$ m 4) $15 - \frac{30}{\pi}$ m

LEVEL-II (ADVANCED)

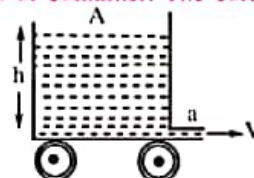
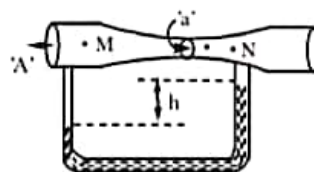
Straight Objective Type Questions

- Blood flows through a main artery of area of cross section 2cm^2 with a speed of 0.25 ms^{-1} . If it branches into several capillaries of total area of cross section 0.4 m^2 , the average speed of blood through each capillary is
 a) 0.25 ms^{-1} b) $1.25 \times 10^{-4}\text{ ms}^{-1}$ c) $12.5 \times 10^{-2}\text{ ms}^{-1}$ d) $0.5 \times 10^{-4}\text{ ms}^{-1}$
- Water flows at the rate of 2ms^{-1} through a pipe of radius 2 cm and emerges from a nozzle of radius 1 cm. If the nozzle is directed vertically upwards, water rises to a maximum height of ($g = 10\text{ ms}^{-2}$)
 a) 3.2 m b) 6.4 m c) 1.6 m d) 0.8 m
- A large vessel with a small hole at the bottom is filled with water and kerosene. The height of the water column is 20 cm and that of the kerosene is 25 cm. The velocity with which water flows out of the hole is (density of kerosene = 0.8 gm/cc) (Neglect viscous force)
 a) 2 ms^{-1} b) 4 ms^{-1} c) 2.8ms^{-1} d) 3 ms^{-1}
- A cylindrical drum open at the top contains 30 litres of water. It drains out through a small opening at the bottom. 10 litres of water comes out in time t_1 , the next 10 litres in further time t_2 and the last 10 litres in further time t_3 . Then
 a) $t_1 = t_2 = t_3$ b) $t_1 > t_2 > t_3$ c) $t_1 < t_2 < t_3$ d) $t_2 > t_1 = t_3$
- A vessel of large uniform cross - sectional area resting on a horizontal surface holds two immiscible, non - viscous and incompressible liquids of densities d and $2d$ each of height $H/2$. If a small hole is punched on the vertical side of the container at a height h ($h < \frac{H}{2}$), the efflux speed of the liquid at the hole is
 a) $\sqrt{2g(H-h)}$ b) $\sqrt{g(H-h)}$ c) $\sqrt{\frac{g}{2}(3H-4h)}$ d) $\sqrt{\frac{g}{2}(4H-3h)}$

6. Water from a tap emerges vertically down with an initial speed of 1.0ms^{-1} . The cross sectional area of tap is $10 \times 10^{-5}\text{m}^2$. Assume that the pressure is constant throughout the stream of water, and that the flow is a steady, the cross sectional area of the stream 0.15m below the tap is
- a) $5.0 \times 10^{-4}\text{m}^2$ b) $1.0 \times 10^{-3}\text{m}^2$ c) $5.0 \times 10^{-3}\text{m}^2$ d) $2.0 \times 10^{-3}\text{m}^2$
7. A large open top container of negligible mass and uniform cross sectional area 'A' has a small hole of cross sectional area 'a' in its side wall near the bottom. The container is kept over a smooth horizontal floor and contains a liquid of density ' ρ ' and mass m_0 . Assuming that the liquid starts flowing through the hole the acceleration of the container will be
- a) $\frac{2ag}{A}$ b) $\frac{ag}{A}$ c) $\frac{2ag}{a}$ d) $\frac{Ag}{a}$
8. A small hole is made at a height of $(1/\sqrt{2})\text{m}$ from the bottom of a cylindrical water tank. The length of the water column is $\sqrt{2}\text{m}$. Find the distance where the water emerging from the hole strikes the ground.
- a) 4m b) 2m c) 3m d) $\sqrt{2}\text{m}$
9. There are two holes one each along the opposite sides of a wide rectangular tank. The cross section of each hole is 0.01m^2 and the vertical distance between the holes is one meter. The tank is filled with water. The net force on the tank in newton when the water flows out of the holes is: (density of water is 1000kg/m^3)
- a) 100 b) 200 c) 300 d) 400
10. Water in a storage tank stands 2.5m above the level of a valve in the side of the tank. With what speed the water will rush out of the valve (neglecting friction)
- a) 5m/s b) 7m/s c) 9m/s d) 11m/s

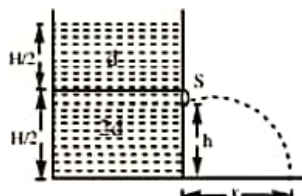
More than One correct answer Type Questions

11. Air is streaming past the wings of an aeroplane with a speed of 90ms^{-1} below and 120ms^{-1} above the surface. If the wing is 15m long and has an average width of 2m , then : [Density of air = 1.2kg/m^3]
- a) Pressure difference between the bottom and top of the wing is 4090Pa
 b) Up – lift on the wing is 81900N
 c) Up – lift on the wing is 113.58kN
 d) Pressure difference between the bottom and top of wing is 3780Pa
12. Water flows through a tube as shows below. Then
- a) h is measure of pressure difference at M and N
 b) Pressure at M > pressure at N
 c) Velocity at N > velocity at M
 d) h is a direct measure of difference in velocity
13. A container having a hole at the bottom is free to move on a frictionless horizontal surface. As the liquid comes out, container moves in backward direction with an acceleration α and finally acquired velocity v (When all liquid drains out) Neglect the mass of container. The correct options are
- a) α is independent of height of liquid in vessel
 b) α depends only A and a and g
 c) v depends upon α and h
 d) v independent of α



Linked Comprehension Type Questions**Passage - I :**

A container of large uniform cross – sectional area A resting on a horizontal surface holds two immiscible, non – viscous and incompressible liquids of densities d and $2d$, each of height $H/2$ as shown. The lower density liquid is open to the atmosphere having pressure P_0 . A tiny hole of area $s(s \ll A)$ is punched on the vertical side of the container at a height $h(h < H/2)$.



14. Initial speed of efflux of the liquid at the hole is

- a) $\sqrt{g(3H-4h)}$ b) $\sqrt{\frac{g}{2}(3H-4h)}$ c) $\sqrt{\frac{g}{3}(3H-4h)}$ d) $\sqrt{\frac{g}{4}(3H-4h)}$

15. Horizontal distance x travelled by the liquid initially is

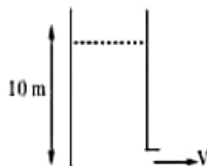
- a) $\sqrt{(3H-4h)h}$ b) $\sqrt{(3H-2h)h}$ c) $\sqrt{(H-h)h}$ d) H

16. The height h at which the hole should be punched so that the liquid travels the maximum distance is

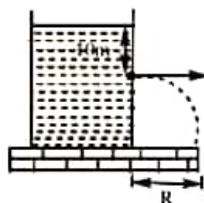
- a) $\frac{H}{4}$ b) $\frac{H}{2}$ c) $\frac{3H}{8}$ d) H

Integer Type Questions

17. An orifice of area 10 sq.mm is made at the bottom of a vessel containing water to a height of 10m above the orifice. What is the force on the vessel in N ($g = 10\text{ms}^{-2}$) ?

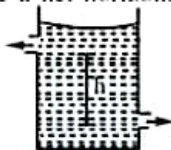


18. A large tank is filled with water. A small hole is made at a depth 10m below water surface. The range of water issuing out of the hole is R on ground. What extra pressure in atm must be applied on the water surface so that the range becomes $2R$: (Take $1\text{atm} = 10^5 \text{ Pa}$ and $g = 10\text{m/s}^2$)



19. A fixed cylindrical vessel is filled with water up to height H . A hole is bored in the wall at a depth h from the free surface of water. For maximum horizontal range ' h ' is equal to H/N where N is equal to

20. There are two identical small holes on the opposite sides of a tank containing a liquid. The tank is open at the top. The difference in height between the two holes is h . As the liquid comes out of the two holes, the tank will experience a net horizontal force proportional to h^n , n is equal to



EXERCISE-IV

(Flow through A Narrow Tube : Poiseuille's Equation, Stokes' Law)

LEVEL-I (MAIN)

Straight Objective Type Questions

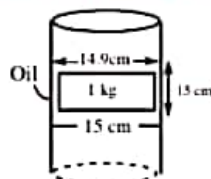
- A liquid is flowing through a narrow tube. The coefficient of viscosity of liquid is 0.1308 poise. The length and inner radius of tube are 50 cm and 1 mm respectively. The rate of flow of liquid is 360 m³/min. Find the pressure difference between ends of tube.
 1) 10^{12} dyne/cm² 2) 10^4 dyne/cm² 3) 10 dyne/cm² 4) 1.8×10^4 dyne/cm²
- Find the terminal velocity of solid sphere of radius 0.1 m moving in air in vertically downward direction. ($\eta = 1.8 \times 10^{-5}$ Ns/m² density of sphere = 1000 kg/m³ and $g = 10$ m/s²)
 1) 2 m/s 2) 2×10^6 cm/s 3) 4 cm/s 4) None of these
- Eight equal drops of water each of radius $r = 2$ mm are falling through air with a terminal velocity of 16 cm/s. The eight drops combine to form a big drop. Calculate the terminal velocity of big drop
 1) 16 m/s 2) 32 cm/s 3) 64 cm/s 4) None of these
- Spherical balls of radius R are falling in a viscous fluid of viscosity η with a velocity v . The retarding viscous force acting on the spherical ball is
 1) directly proportional to R but inversely proportional to v
 2) directly proportional to both radius R and velocity v
 3) inversely proportional to both radius R and velocity v
 4) inversely proportional to R but directly proportional to velocity v .
- Water flows at a speed of 6 cm s⁻¹ through a tube of radius 1 cm. coefficient of viscosity of water at room temperature is 0.01 poise. The Reynolds number is
 1) 100 2) 110 3) 1200 4) 140
- A metal sphere of radius 1 mm and mass 50 mg falls vertically in glycerine, the viscous force exerted by the glycerine on the sphere when the speed of the sphere is 1 m s⁻¹ is (density of glycerine is 1260 kg/m³ coefficient of viscosity at room temperature is 8 poise)
 1) 3×10^{-4} N 2) 150×10^{-4} N 3) 4.5×10^{-4} N 4) 0.5×10^{-4} N
- A narrow tube of radius R and length L is connected in series with another tube of radius $R/2$ and length $L/4$. If the pressure across the two tubes taken together is P , the pressure across the two tubes separately are (Assume steady and stream line flow)
 1) $\frac{4}{5}P$ and $\frac{P}{5}$ 2) $\frac{P}{3}$ and $\frac{2}{3}P$ 3) $\frac{P}{4}$ and $\frac{3}{4}P$ 4) $\frac{P}{2}$ and $\frac{P}{2}$

8. A capillary tube is attached horizontally at a constant head arrangement. If the radius of the capillary tube is increased by 1.5%, the rate of flow of liquid changes by about
- 1) -4% 2) +4% 3) -6% 4) +6%

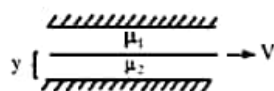
LEVEL-II (ADVANCED)

Straight Objective Type Questions

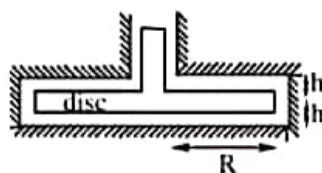
1. A sliding fit cylindrical body of mass 1 kg drops vertically down as shown in figure at a constant velocity of 5 cm/s. Find the coefficient of dynamic viscosity of the oil in pascal second.



- a) 0.2 b) 0.4 c) 0.7 d) 1.4
2. Through a very narrow gap of height h , a thin plate of large extension is pulled. At a velocity v . On one side of the plate there exists oil of viscosity μ_1 and on the other side, oil of viscosity μ_2 as shown. The position (y) of the plate so that (i) the shear force on the two sides of the plate is equal. (ii) the pull required to drag the plate is minimum respectively are



- a) $\frac{\mu_1 h}{\mu_1 + \mu_2} \cdot \frac{h}{1 + \sqrt{\frac{\mu_1}{\mu_2}}}$ b) $\frac{\mu_2 h}{\mu_1 + \mu_2} \cdot \frac{h}{1 + \sqrt{\frac{\mu_1}{\mu_2}}}$ c) $\frac{\mu_1 h}{\mu_1 + \mu_2} \cdot \frac{h}{1 + \sqrt{\frac{\mu_2}{\mu_1}}}$ d) $\frac{\mu_2 h}{\mu_1 + \mu_2} \cdot \frac{h}{1 + \sqrt{\frac{\mu_2}{\mu_1}}}$
3. A thin horizontal disc of radius R is located within a cylindrical cavity filled with oil whose viscosity is η . The distance between the disc and horizontal planes of cavity is h . The power developed by the viscous forces acting on the disc when it rotates with angular velocity ω (The end effects are to be neglected)
- a) $\frac{4\pi\eta\omega^2 R^4}{h}$ b) $\frac{\pi\eta\omega^2 R^4}{4h}$ c) $\frac{\pi\eta\omega^2 R^4}{2h}$ d) $\frac{\pi\eta\omega^2 R^4}{h}$



4. A long cylinder of radius R_1 is displaced along its axis with a contact velocity V_0 inside a stationary co-axial cylinder of radius R_2 . The space between the cylinder is filled with viscous liquid. The velocity of the liquid as a function of the distance r from the axis of the cylinders. The flow is laminar.

- a) $2V_0 \frac{\ln(r/R_2)}{\ln(R_1/R_2)}$ b) $\frac{V_0 \ln(r/R_2)}{2\ln(R_1/R_2)}$ c) $V_0 \frac{\ln(r/R_2)}{\ln(R_1/R_2)}$ d) $\frac{2V_0 \ln(r/R_2)}{3\ln(R_1/R_2)}$

5. Water is flowing on a horizontal fixed surface, such that its flow velocity varies with Y (vertical direction) as $v = k \left(\frac{2y^2}{a^2} - \frac{y^3}{a^3} \right)$. If coefficient of viscosity of water is η . What will be the force per unit area along the layers acting between layers of water at $y = a$.
- a) $\frac{\eta k}{a}$ b) $\frac{2\eta k}{a}$ c) $\frac{\eta a}{k}$ d) $\frac{\eta k}{2a}$
6. 16 cc of water flows per second through a capillary tube of radius a cm and of length 1 cm when connected to a pressure head h cm of water. If a tube of same length and radius $(a/2)$ cm is connected to the same pressure head, the quantity of water flowing through the second tube will be
- a) 16 cc b) 4 cc c) 1 cc d) 8 cc
7. Water in a river 20 m deep is flowing at a speed of 10 ms^{-1} . The shearing stress between the horizontal layers of water in the river in Nm^{-2} is (coefficient of viscosity of water = 10^{-2} SI units)
- a) 2×10^{-2} b) 0.5×10^{-2} c) 1×10^{-3} d) 0.5×10^{-3}
8. Two water pipes of diameter 2 cm and 4 cm are connected to a water tank at the same depth. The length of 4 cm pipe is four times that of the length of 2 cm pipe. The rate of water flow, through 2 cm pipe as compared to that through 4 cm pipe will be
- a) one fourth b) double c) half d) four times

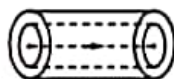
More than One correct answer Type Questions

9. Choose the correct option (s)
- a) with the rise in temperature viscosity decreases
 b) viscosity of liquid increases with increase in pressure but in case of water, viscosity decreases with rise in pressure
 c) viscosity of liquid is about 100 times greater than that of gases
 d) viscosity of gases is independent of pressure
10. The viscous force acting on a solid ball of surface area A moving with terminal velocity v is proportional to
- a) A b) $A^{1/2}$ c) v d) $V^{1/2}$
11. A large wooden plate of area 10 m^2 floating on the surface of a river is made to move horizontally with a speed of 2 m/s by applying a tangential force. River is 1 m deep and the water in contact with the bed is stationary. Then choose correct statement (Coefficient of viscosity of water = $10^{-1} \text{ N} \cdot \text{s}/\text{m}^2$)
- a) velocity gradient is 2 s^{-1} b) velocity gradient is 1 s^{-1}
 c) force required to keep the plate moving with constant speed is 0.02 N
 d) force required to keep the plate moving with constant speed is 0.01 N
12. Choose the correct options
- a) viscosity of liquids increases with temperature
 b) viscosity of gases increases with temperature
 c) surface tension of liquids decreases with temperature
 d) for angle of contact $\theta = 0^\circ$, liquid neither rises nor falls on capillary

Linked Comprehension Type Questions**Passage - I :**

When viscous liquid flows, adjacent layers oppose their relative motion by applying a viscous force given by

$$F = -\eta A \frac{dv}{dy}$$



When η = coefficient of viscosity, A = surface area of adjacent layers in contact, $\frac{dv}{dy}$ = velocity gradient

Now a viscous liquid having coefficient of viscosity η is flowing through a fixed tube of length l and radius R under a pressure difference P between the two ends of the tube.

Consider a cylindrical volume of liquid of radius r . Due to steady flow, net force on the liquid in cylindrical volume should be zero.

$$-\eta 2\pi r l \frac{dv}{dy} = P \pi r^2$$

$$-\int_v^0 dv = \frac{P}{2\eta l} \int_r^R r dr$$

(layer in contact with the tube is stationary)

$$v = v_0 \left(1 - \frac{r^2}{R^2} \right), \text{ where } v_0 = \frac{PR^2}{4\eta l}$$

The volume of the liquid per second through the tube,

$$Q = \int_0^R v \cdot 2\pi r dr = \int_0^R v_0 \left(1 - \frac{r^2}{R^2} \right) 2\pi r dr = v_0 2\pi \left[\frac{r^2}{2} - \frac{r^4}{4R^2} \right]_0^R = v_0 2\pi \left[\frac{R^2}{2} - \frac{R^2}{4} \right] = \frac{v_0 \pi R^2}{2} = \frac{\pi P R^4}{8\eta l}$$

This is called poiseuille's equation.

13. Force acting on the tube due to the liquid is

- a) $\pi\eta/v_0$ b) $2\pi\eta/v_0$ c) $4\pi\eta/v_0$ d) $6\pi\eta/v_0$

14. The viscous force on the cylindrical volume of the liquid varies of

- a) $F \propto r^2$ b) $F \propto r$ c) $F \propto 1/r$ d) $F \propto 1/r^2$

15. The momentum of the liquid confined in the tube is

- a) $\rho\pi R^2/v_0$ b) $\rho\pi R^2/v_0/2$ c) $2\rho\pi R^2/v_0$ d) $\rho\pi R^2/v_0/4$

Matrix Matching Type Questions

16. Two spheres A and B of radii $r/2$, r and densities d , $d/2$, are freely released in a long liquid column of density $d/4$. Consider the stage where all the two attain terminal velocities.

Column - I

- A) Ratio of upthrusts experienced by A, B
B) Ratio of terminal velocities of A,B
C) Ratio of momenta of A,B
D) Ratio of kinetic energies of A,B

Column - II

- P) 3:16
Q) 3:4
R) 9:64
S) 1:8

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

1) 1 2) 3 3) 3 4) 2 5) 1

LEVEL - II

1) a 2) a 3) d 4) b 5) b 6) a 7) b 8) a
 9) abc 10) abcd 11) a 12) c 13) A-Q;B-P;C-S;D-R
 14) 2 15) 8

EXERCISE-II

LEVEL - I

1) 2 2) 2 3) 2 4) 3

LEVEL - II

1) b 2) a 3) b 4) b 5) b 6) c 7) a 8) d
 9) c 10) c 11) abc 12) bcd 13) 2 14) 2

EXERCISE-III

LEVEL - I

1) 1 2) 2 3) 2 4) 1 5) 4

LEVEL - II

1) b 2) a 3) c 4) c 5) c 6) c 7) a 8) d
 9) b 10) b 11) cd 12) abc 13) abc 14) b 15) a 16) c
 17) 2 18) 3 19) 2 20) 1

EXERCISE-IV

LEVEL - I

1) 1 2) 2 3) 3 4) 2 5) 3 6) 2 7) 1 8) 4

LEVEL - II

1) d 2) b 3) d 4) c 5) d 6) c 7) b 8) a
 9) bcd 10) bc 11) ac 12) bc 13) c 14) a
 15) b 16) A-S; B-Q; C-P; D-R